

Euler's Theorem

Statement:

If a and n are positive integers such that

$$\gcd(a, n) = 1$$

then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$ is **Euler's Totient Function** (number of positive integers less than n that are relatively prime to n).

Explanation:

- $\phi(n)$ counts numbers less than n that have no common factor with n .
 - If a is relatively prime to n ,
then raising a to the power $\phi(n)$ gives remainder **1** when divided by n .
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Proof (Simple):

Let the numbers relatively prime to n be:

$$r_1, r_2, r_3, \dots, r_{\phi(n)}$$

Multiply each by a :

$$ar_1, ar_2, ar_3, \dots, ar_{\phi(n)}$$

Since $\gcd(a, n) = 1$, these new numbers are also relatively prime to n , and they are just a rearrangement of the original set.

So,

$$r_1 r_2 \dots r_{\phi(n)} \equiv (ar_1)(ar_2) \dots (ar_{\phi(n)}) \pmod{n}$$

$$r_1 r_2 \dots r_{\phi(n)} \equiv a^{\phi(n)} r_1 r_2 \dots r_{\phi(n)} \pmod{n}$$

Cancel the common term:

$$a^{\phi(n)} \equiv 1(\text{mod } n)$$

Hence proved.

Example:

Verify Euler's Theorem for

$$a = 3, n = 10$$

Step 1: Check gcd

$$\text{gcd}(3, 10) = 1$$

So theorem applies.

Step 2: Find $\phi(10)$

$$10 = 2 \times 5$$

$$\begin{aligned}\phi(10) &= 10 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) \\ &= 10 \times \frac{1}{2} \times \frac{4}{5} \\ &= 4\end{aligned}$$

Step 3: Apply theorem

$$\begin{aligned}3^4 &= 81 \\ 81 \text{ mod } 10 &= 1\end{aligned}$$

✓ Verified

Conclusion:

If $\text{gcd}(a, n) = 1$,

$$\boxed{a^{\phi(n)} \equiv 1(\text{mod } n)}$$

Euler's Theorem – Definition

If a and n are positive integers such that

$$\gcd(a, n) = 1$$

then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where:

- $\phi(n)$ is **Euler's Totient Function**
- $\phi(n)$ = number of positive integers less than n that are relatively prime to n .

In simple words:

If a number a has no common factor with n ,
then a raised to the power $\phi(n)$ leaves remainder **1** when divided by n .